



Web Application Development - BIWA2110

Department	: Faculty of Information Communication Technology	Commence Date	: Week 7
Course Name	: Degree in Business Information Technology/ :Degree in Information Technology	Deadline Date	: Week 10
Semester	: 2	Contact Number	:
		E-mail	:

Individual Assignment 2 15 %

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Assignment Brief

You are required to create a **web-based library management system** for a community library using **React**. The system should save data to **local storage** and allow librarians to manage books, track availability, and handle user accounts.

The application must demonstrate proficiency in:

- React **hooks** (useState, useEffect, etc.)
- **Lifecycle methods** (via hooks)
- **Component composition** (breaking UI into reusable components)
- **Forms** (controlled components, validation, submission handling)
- **React Router** for navigation between pages

Functional Requirements

1. Book Management

- Add new books with: Title, Author, Genre, ISBN, Initial Quantity
- Update existing book details
- Delete books that are no longer available

2. Availability Management

- Track the quantity of each book in stock
- Record transactions:
 - Add stock when new books arrive
 - Deduct stock when books are borrowed

3. User Management

- Login form for users
- Add new users (name, membership ID, role)
- Update user details
- Delete users

User Interface Design

- **Dashboard**
 - Display current book availability in a table or card layout
 - Highlight books with low stock (e.g., fewer than 2 copies)
- **Book Management Page**
 - Form for adding new books
 - Editable list of books with "Update" and "Delete" buttons
- **Transactions Page**
 - Buttons or forms to add/deduct stock
 - Transaction history log
- **User Management Page**
 - Login form
 - Admin view with forms to add/update/delete users

Technology Stack

- **Frontend:** React (JSX, Components, Hooks, Router)
- **Styling:** CSS (or styled-components if preferred)
- **State Management:** React hooks (useState, useEffect)
- **Database:** Local Storage

Marking Criteria

User Interface (consistent, responsiveness)	20
Functionality	25

Code Quality (React best practices, component structure)	15
Student's Own Work	20
Total	80

SUBMISSION INFORMATION

Submit a link to GitHub project